
NUMERICAL COMPUTATION OF HADAMARD FINITE PART OF SOME HYPERSINGULAR INTEGRALS IN 1, 2 AND 3 DIMENSIONS

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ABSTRACT

Data mining is a good way to find the relationship between raw data and predict the target we want which is also widely used in different field nowadays. In this project, we implement a lots of technology and method in data mining to predict the sale of an item based on its previous sale. We create a strong model to predict the sales. After evaluating this model, we conclude that this model can be used in normal life for future sale's prediction.

1 Introduction

Our project is a competition on Kaggle (Predict Future Sales). We are provided with daily historical sales data (including each products' sale date, block ,shop price and amount). And we will use it to forecast the total amount of each product sold next month. Because of the list of shops and products slightly changes every month. We need to create a robust model that can handle such situations.

2 Notations

For every physical quantity, the corresponding reference space quantity is denoted by a hat, so that, for example, if $\mathbf{x}$ is a point on a d -dimensional variety, then $\hat{\mathbf{x}}$ is the corresponding point in the reference space. If u is a function defined on the variety, then $\hat{u}$ is the corresponding function in the reference space. Furthermore, we denote by $\oint$ the Hadamard finite part of an integral.

3 Model presentation

Let's consider an finite part integral of the form

$$I = \oint_{\hat{\tau}} f(\hat{\mathbf{y}}) d\hat{\mathbf{y}}, \quad f(\hat{\mathbf{y}}) \underset{\hat{\mathbf{y}} \rightarrow \hat{\mathbf{x}}}{\sim} \frac{1}{\|\hat{\mathbf{y}} - \hat{\mathbf{x}}\|^{d+1}}. \quad (1)$$

4 Injecting Richardson extrapolation into Guiggiani's final formula

4.1 Richardson extrapolation

For any angle θ , let's consider a finite sequence of n radial points $\rho_i(\theta)$ such that

$$0 < \rho_1(\theta) < \rho_2(\theta) < \dots < \rho_n(\theta) < \hat{\rho}(\theta), \quad (2)$$

Recall: we can express the two first terms of the Laurent expansion of the complete polar kernel $F(\rho, \theta) = \rho K(\chi(\hat{\mathbf{x}}), \chi(\hat{\mathbf{y}}(\rho, \theta)))J(\hat{\mathbf{y}}(\rho, \theta))\hat{u}(\hat{\mathbf{y}}(\rho, \theta))$ as the following limits

$$F_{-2}(\theta) = \lim_{\rho \rightarrow 0} \rho^2 F(\rho, \theta), \quad F_{-1}(\theta) = \lim_{\rho \rightarrow 0} \rho \left(F(\rho, \theta) - \frac{F_{-2}(\theta)}{\rho^2} \right). \quad (3)$$

such that we can write the complete polar kernel as

$$F(\rho, \theta) = \frac{F_{-2}(\theta)}{\rho^2} + \frac{F_{-1}(\theta)}{\rho} + O_{\rho \rightarrow 1}(1) \quad (4)$$

Let $c_{1,i}$ and $c_{2,i}$ be the coefficients used in the Richardson extrapolation of order n to compute the previous limits, so that we can write

$$\boxed{F_{-2}(\theta) = \sum_{i=1}^n c_{2,i} \rho_i^2(\theta) F(\rho_i(\theta), \theta)}, \quad F_{-1}(\theta) = \sum_{i=1}^n c_{1,i} \rho_i(\theta) \left(F(\rho_i(\theta), \theta) - \frac{F_{-2}(\theta)}{\rho_i^2(\theta)} \right). \quad (5)$$

By injecting the formula of $F_{-2}(\theta)$ into the formula of $F_{-1}(\theta)$, so that we can write $F_{-1}(\theta)$ only as a linear combination of the complete polar kernel evaluated at the radial points $\rho_i(\theta)$, we obtain

$$F_{-1}(\theta) = \sum_{i=1}^n c_{1,i} \rho_i(\theta) F(\rho_i(\theta), \theta) - \sum_{i=1}^n c_{1,i} \frac{1}{\rho_i(\theta)} \sum_{j=1}^n c_{2,j} \rho_j^2(\theta) F(\rho_j(\theta), \theta).$$

By letting $P_n(\theta) = \sum_{k=1}^n \frac{1}{\rho_k(\theta)}$ be a constant that is a polynomial in $\cos \theta$ and $\sin \theta$, we can write the previous formula as

$$\boxed{F_{-1}(\theta) = \sum_{i=1}^n F(\rho_i(\theta), \theta) \rho_i(\theta) \left(c_{1,i} - c_{2,i} P_n(\theta) \rho_i(\theta) \right)} \quad (6)$$

4.2 Final quadrature rule

Let's then denote $I(\theta)$ the integral of the complete polar kernel over the radial direction, defined as:

$$I(\theta) = \int_0^{\hat{\rho}(\theta)} \left\{ F(\rho, \theta) - \frac{F_{-1}(\theta)}{\rho} - \frac{F_{-2}(\theta)}{\rho^2} \right\} d\rho + F_{-1}(\theta) \ln \left| \frac{\hat{\rho}(\theta)}{\beta(\theta)} \right| - F_{-2}(\theta) \left\{ \frac{\gamma(\theta)}{\beta^2(\theta)} + \frac{1}{\hat{\rho}(\theta)} \right\} \quad (7)$$

A final quadrature rule for it is the following:

$$I(\theta) \approx \tilde{I}(\theta) = \sum_{i=1}^n \tilde{w}_i(\theta) F(\rho_i(\theta), \theta) \quad (8)$$

With the new weights $\tilde{w}_i(\theta)$ defined as

$$\begin{aligned} \tilde{w}_i(\theta) = & w_i(\theta) - c_{1,i}\rho_i(\theta)W_{-1} + c_{2,i}\rho_i^2(\theta)(P_n(\theta)W_{-1} - W_{-2}(\theta)) + \\ & \rho_i(\theta) \left\{ \ln \left| \frac{\hat{\rho}(\theta)}{\beta(\theta)} \right| (c_{1,i} - P_n(\theta)c_{2,i}\rho_i(\theta)) - \left(\frac{\gamma(\theta)}{\beta^2(\theta)} + \frac{1}{\hat{\rho}(\theta)} \right) c_{2,i}\rho_i(\theta) \right\} \end{aligned}$$

and the auxiliary constants :

- $w_i(\theta) = \frac{\hat{\rho}(\theta)}{2} w_i^{\text{Ref}}$ are the weights of the reference quadrature over the segment $[-1, 1]$.
- $W_{-1} = \sum_{i=1}^n \frac{w_i(\theta)}{\rho_i(\theta)} = \sum_{i=1}^n \frac{w_i^{\text{Ref}}}{1+x_i^{\text{Ref}}}.$
- $W_{-2}(\theta) = \sum_{i=1}^n \frac{w_i(\theta)}{\rho_i^2(\theta)} = \frac{2}{\hat{\rho}(\theta)} \sum_{i=1}^n \frac{w_i^{\text{Ref}}}{(1+x_i^{\text{Ref}})^2}.$

Remark: in the case of Gauss-Legendre quadrature, the constants W_{-1} and W_{-2} can be computed analytically as:

$$W_{-1} = 2H_n, \quad W_{-2}(\theta) = \frac{2}{\hat{\rho}(\theta)} n(n+1) \quad (9)$$

With $H_n = \sum_{i=1}^n \frac{1}{i}$ the n -th harmonic number.